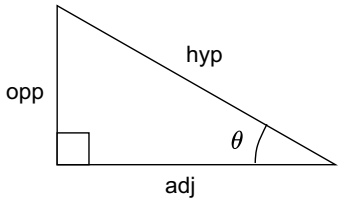
1.39 Trigonometry: Introduction

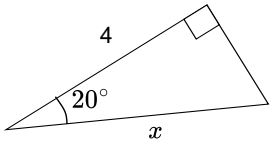
Chapter 9 · Original slide 4

RECAP: Right-Angled Trigonometry

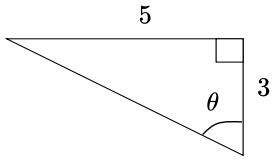
4



You are probably familiar with the formula: $\sin(\theta) = \frac{opp}{hyp}$
But what is the *conceptual* definition of $\sin$?
sin is a function which inputs an angle and gives the ratio between the opposite and hypotenuse.



Find x.



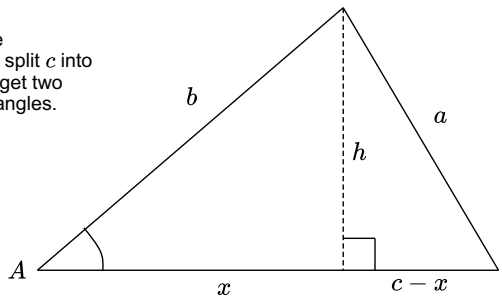
Find θ.

Chapter 9 · Original slide 5

Proof of Cosine Rule

5

We want to use Pythagoras, so split c into two so that we get two right-angled triangles.



By Pythagoras: $h^2 + x^2 = b^2$ and $h^2 + (c-x)^2 = a^2$
Subtracting to eliminate h:
$x^2 - (c-x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$
But $x = b\cos A$ $\therefore$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

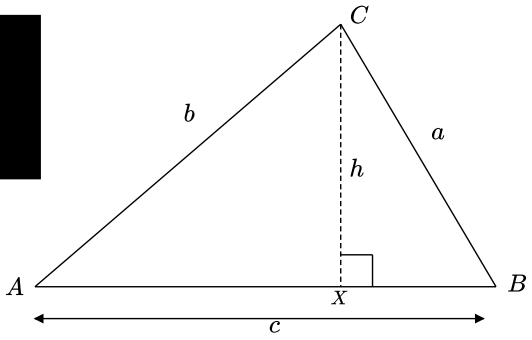
1.39 Trigonometry: Introduction

Chapter 9 · Original slide 6

Proof of Sine Rule

6

The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.



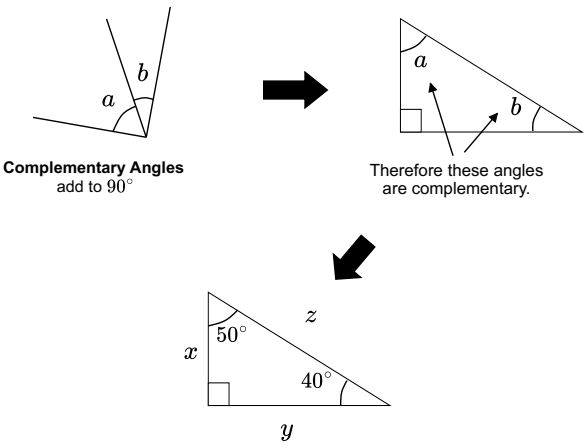
Using basic trigonometry:
$h = a\sin B$ and $h = b\sin A$
$\therefore a\sin B = b\sin A$
$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$

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Just for your interest…

7

Why does “cosine” contains the word “sine”?



1.39 Trigonometry: Introduction

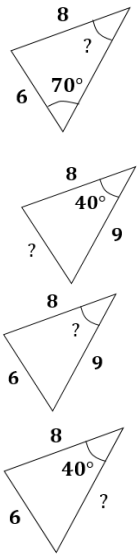
Chapter 9 · Original slide 8

OVERVIEW: Finding missing sides and angles

8

When triangles are not right-angled, we can no longer use simple trigonometric ratios, and must use the cosine and sine rules.

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	
#2 Two sides known and a missing side opposite a known angle	Remaining side	
#3 All three sides	An angle	
#4 Two sides known and a missing side not opposite known angle	Remaining side	

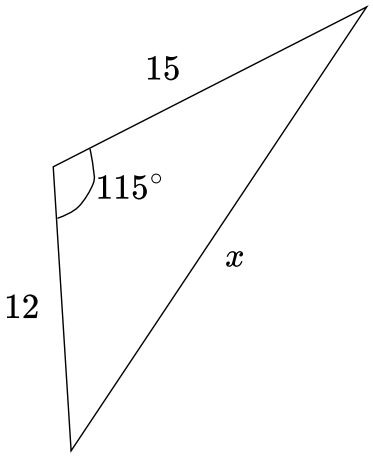


Chapter 9 · Original slide 9

Cosine Rule

9

We use the **cosine rule** whenever we have **three sides** (and an angle) involved.



Proof at end of PowerPoint.

Cosine Rule:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

The only angle in formula is A, so label angle in diagram A, label opposite side a, and so on (b and c can go either way).

1.39 Trigonometry: Introduction

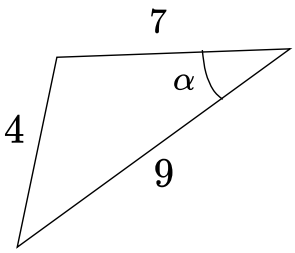
Chapter 9 · Original slide 10

Dealing with Missing Angles

10

You have	You want	Use
All three sides	An angle	Cosine rule

$$\mathbf{a^2=b^2+c^2-2bccosA}$$



Label sides then substitute into formula.

Simplify each bit of formula.

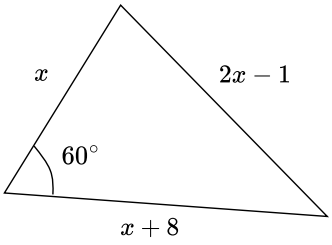
Rearrange "swapping" thing you're subtracting and result

Chapter 9 · Original slide 11

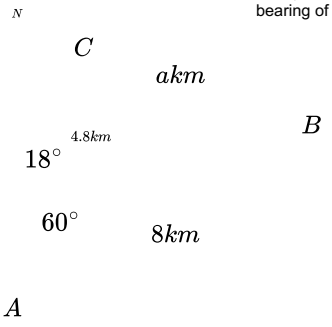
Harder Ones

11

Determine the value of x.



[From textbook] Coastguard station B is 8 km, on a bearing of 060°, from coastguard station A. A ship C is 4.8 km on a bearing of 018°, away from A. Calculate how far C is from B.



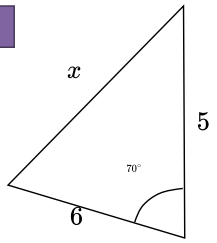
1.39 Trigonometry: Introduction

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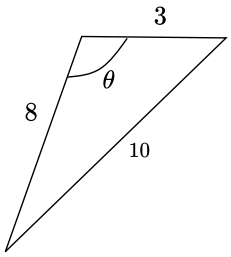
Test Your Understanding

12

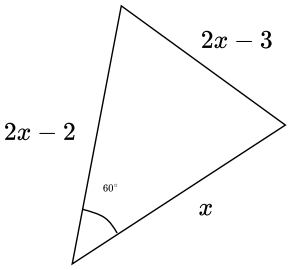
1



2



3

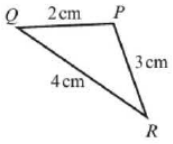


Chapter 9 · Original slide 13

Exercise 9A – Y1 pages 177-179

13

8 Show that $\cos P = -\frac{1}{4}$.



11 The lengths of the sides of a triangle are in the ratio $2 : 3 : 4$.
Calculate the size of the largest angle.

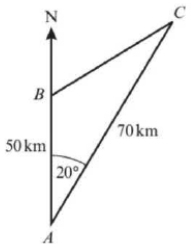
1.39 Trigonometry: Introduction

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Exercise 9A – Y1 pages 177-179

14

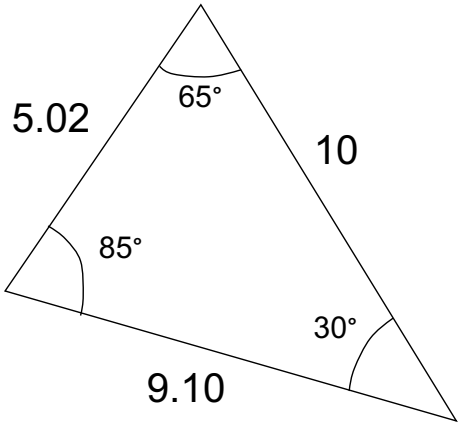
17 The diagram shows three cargo ships, A, B and C, which are in the same horizontal plane. Ship B is 50 km due north of ship A and ship C is 70 km from ship A. The bearing of C from A is $020°$.
a Calculate the distance between ships B and C, in kilometres to 3 s.f. (3 marks)
b Calculate the bearing of ship C from ship B. (4 marks)



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The Sine Rule

16



For this triangle, try calculating each side divided by the sin of its opposite angle. What do you notice in all three cases?

Sine Rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

You have	You want	Use
#1. Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

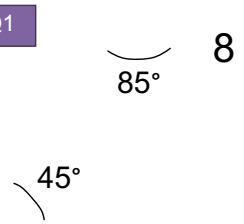
1.39 Trigonometry: Introduction

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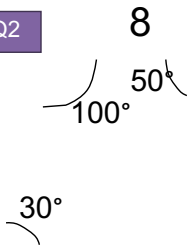
Examples

17

Q1



Q2



You have	You want	Use
#1. Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

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Examples

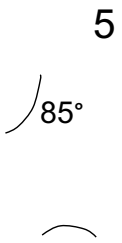
18

When you have a missing angle, it's better to reciprocate to get:

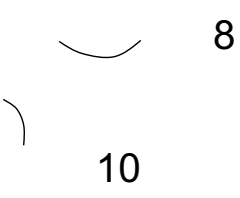
$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

i.e. in general put the missing value in the numerator.

Q3



Q4



1.39 Trigonometry: Introduction

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Exercise 9B – Y1 pages 181-183

19

8 Town B is 6 km, on a bearing of 020°, from town A. Town C is located on a bearing of 055° from town A and on a bearing of 120° from town B. Work out the distance of town C from:
a town A\qquad b town B

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Exercise 9B – Y1 pages 181-183

20

11 In $\triangle ABC$,
$AB = x$ cm, $BC = (4 - x)$ cm, $\angle BAC = y$ and
$\angle BCA = 30^\circ$.
Given that $\sin y = \frac{1}{\sqrt{2}}$, show that

$$x = 4(\sqrt{2} - 1)$$

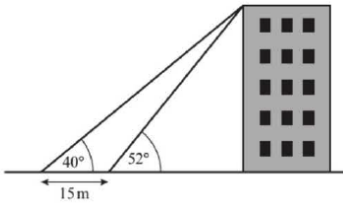
1.39 Trigonometry: Introduction

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Exercise 9B – Y1 pages 181-183

21

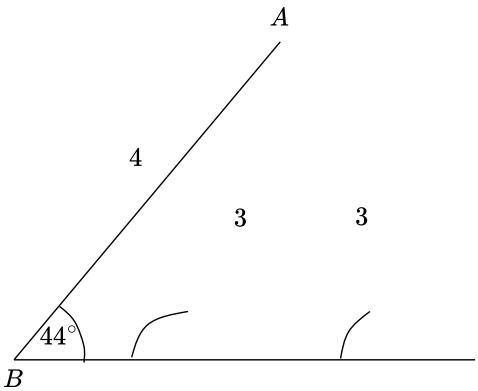
12 A surveyor wants to determine the height of a building. She measures the angle of elevation of the top of the building at two points 15 m apart on the ground.
a Use this information to determine the height of the building. (4 marks)
b State one assumption made by the surveyor in using this mathematical model. (1 mark)



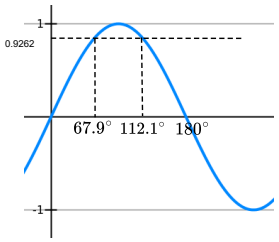
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The ‘Ambiguous Case’

23



Suppose you are told that $AB = 4$, $AC = 3$ and $\angle ABC = 44^\circ$. What are the possible values of $\angle ACB$?



Your calculator will give the acute angle of 67.9° (i.e. C_2). But if we look at a graph of sin, we can see there’s actually a second value for $\sin^{-1}(0.9262)$, corresponding to angle C_1.

The sine rule produces two possible solutions for a missing angle:

$$\sin\theta = \sin(180^\circ - \theta)$$

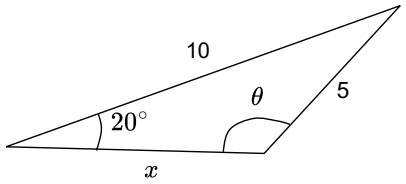
Whether we use the acute or obtuse angle depends on context.

1.39 Trigonometry: Introduction

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Test Your Understanding

24



Given that the angle θ is obtuse, determine θ and hence determine the length of x.

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Exercise 9C – Y1 pages 184-185

25

4 Triangle ABC is such that $AB = 4$ cm, $BC = 6$ cm and $\angle ACB = 36°$. Show that one of the possible values of $\angle ABC$ is $25.8°$ (to 3 s.f.). Using this value, calculate the length of AC.

1.39 Trigonometry: Introduction

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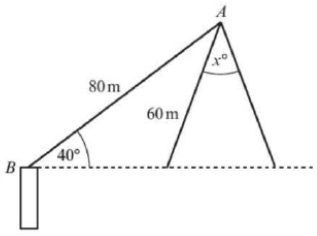
Exercise 9C – Y1 pages 184-185

5 Two triangles ABC are such that $AB = 4.5\text{ cm}$, $BC = 6.8\text{ cm}$ and $\angle ACB = 30^\circ$. Work out the value of the largest angle in each of the triangles.

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Exercise 9C – Y1 pages 184-185

6 a A crane arm AB of length 80 m is anchored at point B at an angle of 40° to the horizontal. A wrecking ball is suspended on a cable of length 60 m from A. Find the angle x through which the wrecking ball rotates as it passes the two points level with the base of the crane arm at B. (6 marks)
b Write down one modelling assumption you have made. (1 mark)

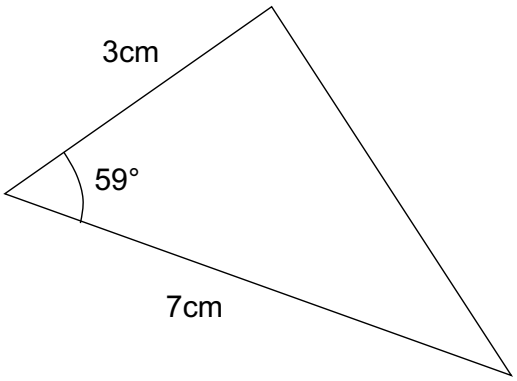


1.39 Trigonometry: Introduction

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Area of Non Right-Angled Triangles

28



$$\text{Area} = \frac{1}{2}ab\sin(C)$$

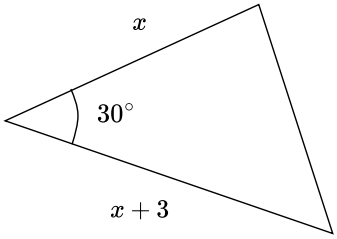
where C is the angle between two sides a and b.

Tip: You shouldn't have to label sides/angles before using the formula. Just remember that the angle is between the two sides.

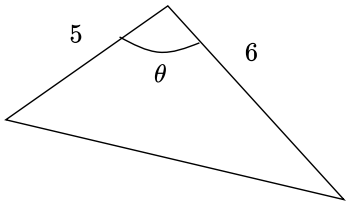
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Test Your Understanding

29



The area of this triangle is 10.
Determine x.



The area of this triangle is also 10.
If θ is obtuse, determine θ.

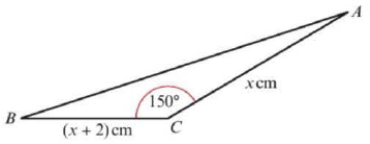
1.39 Trigonometry: Introduction

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Exercise 9D – Y1 pages 186-187

30

4 In triangle ABC, $BC = (x + 2)$ cm, $AC = x$ cm and $\angle BCA = 150^\circ$. Given that the area of the triangle is 5 cm^2, work out the value of x, giving your answer to 3 significant figures.



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Exercise 9D – Y1 pages 186-187

31

5 In $\triangle PQR$, $PQ = (x + 2)$ cm, $PR = (5 - x)$ cm and $\angle QPR = 30^\circ$. The area of the triangle is A cm^2.

a Show that $A = \frac{1}{4}(10 + 3x - x^2)$.

b Use the method of completing the square, or otherwise, to find the maximum value of A, and give the corresponding value of x.

1.39 Trigonometry: Introduction

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Exercise 9D – Y1 pages 186-187

32

6 In $\triangle ABC$, $AB = x$ cm, $AC = (5 + x)$ cm and $\angle BA$
that the area of the triangle is $3\frac{3}{4}$ cm^2
a Show that x satisfies the equation $x^2 + 5x - 15 = 0$.
b Calculate the value of x, giving your answer to 3 significa... figures.

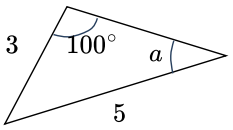
Chapter 9 · Original slide 33

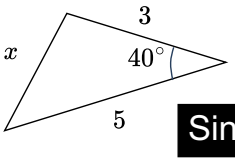
Sin or cosine rule?

33

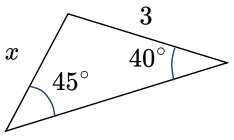
Recall that whenever we have **two "side-angle pairs"** involved, use sine rule. If there's **3 sides** involved, we can use cosine rule. Sine rule is generally easier to use than cosine rule.

Sine Cosine

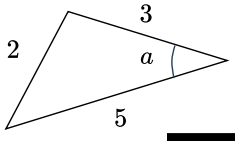




Sine Cosine



Sine Cosine



Sine Cosine

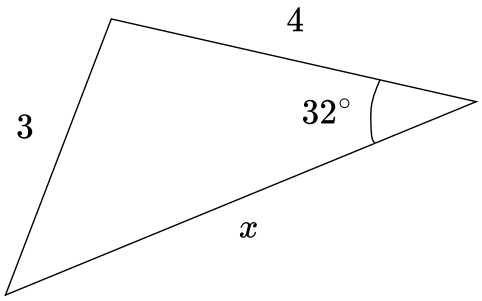
1.39 Trigonometry: Introduction

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Using sine rule twice

34

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice



Given there is just one angle involved, you might attempt to use the cosine rule:

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Using sine rule twice

35

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice

1: We could use the sine rule to find this angle.

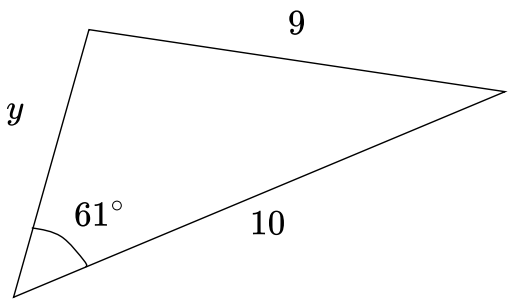
$$\frac{4}{\sin 32^\circ} = \frac{3}{\sin x}$$

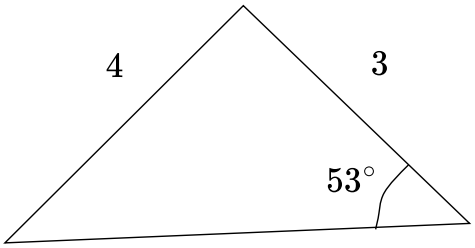
1.39 Trigonometry: Introduction

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Test Your Understanding

36



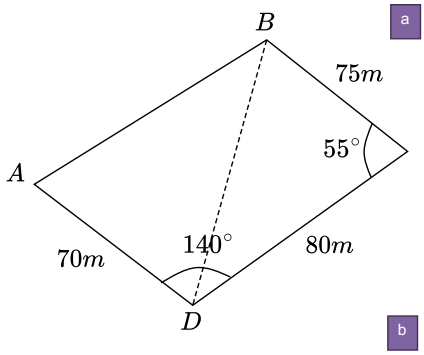


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Problem Solving With Sine/Cosine Rule

37

[From Textbook] The diagram shows the locations of four mobile phone masts in a field, $BC = 75m$. $CD = 80m$, angle $BCD = 55^\circ$ and angle $ADC = 140^\circ$.
In order that the masts do not interfere with each other, they must be at least 70m apart.
Given that A is the minimum distance from D, find:
a) The distance A is from B
b) The angle BAD
c) The area enclosed by the four masts.



a

b

c

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12 The perimeter of $\triangle ABC = 15\text{ cm}$. Given that $AB = 7\text{ cm}$ and $\angle BAC = 60°$, find the lengths of AC and BC and the area of the triangle.

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13 In the triangle ABC, $AB = 14\text{ cm}$, $BC = 12\text{ cm}$ and $CA = 15\text{ cm}$.
a Find the size of angle C, giving your answer to 3 s.f.
b Find the area of triangle ABC, giving your answer in cm^2 to 3 s.f.

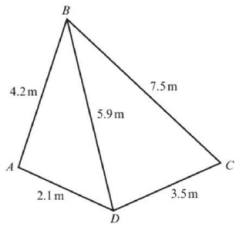
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14 A flower bed is in the shape of a quadrilateral as shown in the diagram.
a Find the sizes of angles DAB and BCD. \hfill(4 ma...
b Find the total area of the flower bed. \hfill(3 ma...
c Find the length of the diagonal AC.



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15 $ABCD$ is a square. Angle CED is obtuse.
Find the area of the shaded triangle. (7 marks)

